Relational Clocks and Causal Cartan Contraction through Arity Three
in Pure-Weyl Gravity on a Berger Universe

GPT-5.6.sol*

Working draft, July 2026

Abstract

We study whether cylinder-time translation can remain presymplectically null after a healthy matter clock and the first nonlinear vertices are included in pure-Weyl gravity. On $\mathbb{R} \times S^3$ with a Berger spatial metric we exhibit an open family of exact, smooth, non-conformally-flat solutions sourced by two standard-sign conformal scalars. Their rotating phase is an everywhere timelike clock, the quartic potential is positive, and the internal clock momentum is nonzero. Nevertheless, at fixed couplings the linearized lapse constraint and compact spatial averaging force its variation to vanish on every smooth solution tangent. The covariant identity $\Omega_{\text{total}}(\delta, \mathcal{L}_D) = \omega\, \delta Q_R$ therefore makes the cylinder generator D a presymplectic degeneracy on this declared linearized phase space. This proves infinitesimal nullity, not integrability of a nonlinear quotient.

At one exact rational member we then formulate the same conclusion homologically. The complete gauge-fixed classical BV presentation has 54 rows and a support-local cyclic retract onto a 26-row retained complex. Advanced and retarded chain contractions lift to all rows. Differentiating the covariant master action gives complete arbitrary-input four-dimensional operations q_2 and q_3, with the required L_∞ identities, cyclicity and local D-equivariance. The unary and binary Cartan primitives extend to a cyclic ternary primitive: the arity-three source closes by the graded Jacobi identity, and a causal raw primitive becomes cyclic under the exact C_4 Reynolds projector. Thus the D action is causally null-homotopic through arity three. An independent reduced-action calculation crosses the producer boundary in eight lapse and radial/Weyl fourth derivatives, agreeing with sixteen ordered coefficients of the frozen q_3 payload; it is not a second full derivation.

The result is classical: its linear nullity statement holds on the displayed open fixed-coupling Berger branch, while its all-row BV Taylor statement is proved at one exact rational representative. It is not an all-orders nonlinear theorem, a Hadamard construction, a quantum-master-equation result, an anomaly calculation, or a claim that a nonzero clock momentum is a freely variable boundary charge.

Authorship and computational inputs

The named model author produced the mathematical programme, derivations, verification code, referee responses, and manuscript. Asger Alstrup Palm commissioned and directed the work, initiated the questions, and served as non-technical orchestrator and corresponding human contact;

*OpenAI model. Contributed the research programme, mathematical direction, derivations, machine verification, referee analyses, and manuscript. The project was commissioned and directed by Asger Alstrup Palm (`asger@area9.dk`), who initiated the questions, served as the non-technical orchestrator and corresponding human contact, but claims no technical contribution.

he claims no technical contribution. Repository scripts generate the certificates and sparse coefficient tables used as computational inputs. They are not additional authors. This article has no generated TEX input. This attribution describes the research archive. Any submission to a venue with a different authorship policy requires a separate accountability review; that review is not represented by the present working draft.

Contents

1 Question, phase space and theorem boundary

There are two superficially conflicting requirements for a relational clock. Its phase must evolve and possess a nonzero conjugate momentum, yet the total Hamiltonian generator should remain a constraint on a closed universe. A matter phase with zero momentum is not a clock, while a freely variable nonzero Hamiltonian charge would make time translation a physical symmetry rather than proper gauge. The distinction is made by the pullback of the charge variation to the allowed solution space, not by the background value of the charge. We use the BV description of gauge systems [2, 3], the covariant phase-space pairing [7], and causal chain contractions in the sense of Green-hyperbolic complexes [16].

We work on

$$M = \mathbb{R} \times S^3 \tag{1}$$

with no spatial boundary. Couplings are fixed. Write $\mathcal{E}_{\alpha_B,\lambda}(\Phi) = 0$ for the covariant equations and $\bar{\Phi}$ for the Berger solution. The charge theorem is evaluated on the pre-reduced smooth tangent space

$$\mathcal{Z}_{\bar{\Phi}}^{\alpha_B,\lambda} := \ker D\mathcal{E}_{\alpha_B,\lambda}\big|_{\bar{\Phi}} \subset \Gamma^\infty(M,F), \qquad \delta\alpha_B = \delta\lambda = 0, \tag{2}$$

including the linearized constraints. We test whether the D vector lies in the radical of the covariant presymplectic form on this space; we do not quotient by D in the definition. The nonlinear statement is instead a formal Taylor identity for the classical BV vector field through arity three at the same background. These are related statements, but they are not identical.

Let $D := \partial_t$ denote cylinder-time diffeomorphism and let R generate the global, ungauged internal $O(2)$ rotation of the scalar doublet. On the background,

$$\mathcal{L}_D\bar{g} = 0, \qquad \mathcal{L}_D\bar{T} = \omega R\bar{T}, \qquad K := D - \omega R, qquad\mathcal{L}_K\bar{\Phi} = 0. \tag{3}$$

Thus the solution is stationary in stress energy but not pointwise fixed by D on the scalar fields. The orbit of D is helical in spacetime times the internal target. The paper tests the diffeomorphism generator D, not the stabilizer K and not the global rotation R.

Theorem 1.1 (Invariant scoped Berger theorem)**.** *Fix $\alpha_B > 0$. For every point of the positive Berger branch described in Section 2, assertions (i)–(ii) below hold. At the exact rational representative* (17)*, assertions (iii)–(iv) also hold:*

1. *the background carries a smooth standard-sign conformal-scalar phase clock with positive quartic potential, timelike phase gradient and nonzero internal momentum;*
2. *on the smooth fixed-coupling linearized solution tangent $\mathcal{Z}_{\bar{\Phi}}^{\alpha_B,\lambda}$,*

$$\Omega_{\text{total}}(\delta, \mathcal{L}_D) = 0 \tag{4}$$

   for every allowed tangent δ;

3. *the complete 54-row gauge-fixed classical BV complex admits advanced and retarded chain contractions with causal support;*
4. *if q_1, q_2, q_3 are the first three action-derived Taylor operations, then there are cyclic causal Cartan components $\iota_D^{(1)}, \iota_D^{(2)}, \iota_D^{(3)}$ satisfying the Cartan equation through arity three; equivalently,*

$$\big([Q, \iota_D] - L_D\big)^{(n)} = 0, \qquad n = 1, 2, 3, \tag{5}$$

   with an uncomputed arity-four-and-higher remainder.

Consequently D is presymplectically null at the linearized covariant level and is causally null-homotopic through the first ternary BV interaction test. No constant-rank neighbourhood, integrated null foliation, finite nonlinear quotient or all-orders moment-map theorem is asserted.

The four assertions are certified respectively by `P09-C1`–`P09-C2`, `P09-C3`–`P09-C4`, `P09-C5`–`P09-C6`, and `P09-C7`–`P09-C10` in the companion computational supplement. The theorem does not assert that the Cartan contraction extends to all Taylor orders or to a renormalized quantum theory.

Proof architecture. The geometric layer derives the exact background and charge identity from the action. The analytic layer constructs the retained Green homotopies and lifts them through a support-local deformation retract. The algebraic layer differentiates the master action and solves the first three Cartan recurrences. Computer algebra certifies the instantiated curved coefficients and all-row signs; it does not replace the invariant definitions or the compact-averaging argument.

Novelty and relation to established machinery. Relational and complete observables, covariant phase-space charges, linearization stability, BV–BFV theory, cyclic homotopy algebras, homotopy transfer, Cartan homotopies and Green homotopies are established frameworks [4, 5, 6, 8, 10, 11, 12, 13, 14, 16]. The contribution claimed here is their explicit conjunction in one action-derived example: a positive fully backreacting Berger clock, a fixed-coupling presymplectic-nullity theorem, a complete gauge-fixed classical BV presentation, and a cyclic causally supported Cartan primitive through q_3. To our knowledge the specific Berger/Weyl/scalar construction is new; no priority claim is made for any of the general frameworks.

2 A positive clock on an exact Berger background

Let σ_i be a left-invariant coframe on S^3 normalized by

$$d\sigma_1 = -\sigma_2 \wedge \sigma_3, qquad d\sigma_2 = -\sigma_3 \wedge \sigma_1, qquad d\sigma_3 = -\sigma_1 \wedge \sigma_2, qquad \int_{S^3} \sigma_1 \wedge \sigma_2 \wedge \sigma_3 = 16\pi^2. \quad (6)$$

We use metric signature $(-,+,+,+)$. The curvature convention is fixed operationally by

$$\delta \int \sqrt{-g}\, C^2 = 4 \int \sqrt{-g}\, B_{\mu\nu}\, \delta g^{\mu\nu}, \quad (7)$$

which, with the matter stress convention below, gives $\alpha_B B_{\mu\nu} = T_{\mu\nu}$. Set

$$g = -dt^2 + a^2(\sigma_1^2 + \sigma_2^2) + c^2\sigma_3^2, \qquad q = \frac{c^2}{a^2}. \quad (8)$$

The matter fields are two real conformal scalars with positive target metric, written in polar form as

$$T_1 = \rho \cos(\omega t), \qquad T_2 = \rho \sin(\omega t). \quad (9)$$

Their phase $\theta = \omega t \pmod{2\pi}$ obeys $g^{-1}(d\theta, d\theta) = -\omega^2 < 0$ and is therefore an everywhere timelike S^1-valued clock candidate. The temporal-diffeomorphism/Weyl incidence determinant is $\rho^2\omega$, so the clock slice is regular on the branch below.

Our action convention is

$$S = \int_M \sqrt{-g} \left[\frac{\alpha_B}{8} C_{\mu\nu\rho\sigma} C^{\mu\nu\rho\sigma} - \frac{1}{2} \sum_{A=1}^{2} \nabla_\mu T_A \nabla^\mu T_A - \frac{R}{12}\rho^2 - \frac{\lambda}{4}\rho^4 \right]. \quad (10)$$

With $T_{\mu\nu} = -2\delta S_m/\delta g^{\mu\nu}$, its metric equation is

$$\alpha_B B_{\mu\nu} = T_{\mu\nu}. \quad (11)$$

Proposition 2.1 (Exact positive Berger family; `P09-C1`, `P09-C2`)**.** *For*

$$\frac{5 - \sqrt{21}}{2} < q < \frac{1}{4} \quad (12)$$

there is an exact solution of the scalar and metric equations with

$$\rho^2 = \frac{2\alpha_B(1-4q)}{a^2}, \qquad \omega^2 = \frac{q}{4a^2(1-4q)}, \qquad \lambda = -\frac{q^2 - 5q + 1}{6\alpha_B(1-4q)^2} > 0. \quad (13)$$

The scalar kinetic metric is positive, the quartic potential is bounded below, and the exact stress obeys the pointwise dominant-energy inequalities $0 \le p_h, p_v \le \varepsilon$ in the orthonormal frame.

Proof. In the orthonormal Berger frame the independent Bach components are

$$B_{00} = \frac{(a^2 - c^2)^2}{6a^8},$$

$$B_{11} = B_{22} = \frac{(a^2 - c^2)(a^2 - 3c^2)}{6a^8}, \quad (14)$$

$$B_{33} = \frac{(a^2 - c^2)(5c^2 - a^2)}{6a^8}. \quad (15)$$

Substitution of (9) into the scalar equation and the three independent components of (11) gives (13). The interval (12) makes ρ^2, ω^2 and λ positive. Direct substitution gives

$$\varepsilon = \frac{\rho^2(1-q)^2}{12a^2(1-4q)},$$
$$p_h = \frac{\rho^2(1-q)(1-3q)}{12a^2(1-4q)}, \qquad p_v = \frac{\rho^2(1-q)(5q-1)}{12a^2(1-4q)}, \tag{16}$$

from which $0 \le p_h, p_v \le \varepsilon$ follows on the stated interval. The coefficientwise verification also checks the lapse, horizontal-scale and vertical-scale variations of the Weyl-square action. □

These properties define the word "healthy" in this paper: standard-sign target kinetic form, nonzero amplitude, timelike phase gradient, positive bounded-below potential on the solution, and the displayed pointwise dominant-energy inequalities. They are properties of this exact solution, not a claim that the improved conformal-scalar stress satisfies an energy condition on arbitrary configurations [15]. The inequalities are pointwise and on shell in the displayed Weyl representative; they are not asserted to be invariant under changing that representative.

Remark 2.2 (S^1 clock and local real lifts). Because $\rho > 0$, the map $(T_1, T_2)/\rho$ is globally S^1-valued and has period $2\pi/\omega$ along the D orbit. A real clock coordinate exists only after choosing a local lift of θ on an interval shorter than one period. The slice $\theta = \theta_0$ therefore meets a complete orbit periodically. We prove transversality of this local clock slice, not a global one-intersection deparametrization and not absence of Gribov repetitions.

The rational point

$$a = 1, \quad q = \frac{9}{40}, \quad \alpha_B = 5, \quad \rho = 1, \quad \omega = \frac{3}{4}, \quad \lambda = \frac{119}{480} \tag{17}$$

is used for the exact all-row BV computations. It is a representative of the open branch, not a finite-mode replacement for the four-dimensional operations.

3 Nonzero clock momentum and vanishing tangent charge

The matter action has an internal $O(2)$ rotation

$$R(T_1, T_2) = (-T_2, T_1) \tag{18}$$

with current

$$j_R^\mu = T_1 \nabla^\mu T_2 - T_2 \nabla^\mu T_1. \tag{19}$$

For the Maurer–Cartan volume convention $\int_{S^3} \sigma_1 \wedge \sigma_2 \wedge \sigma_3 = 16\pi^2$, the background charge is

$$Q_R = 16\pi^2 \alpha_B q \sqrt{1-4q} > 0. \tag{20}$$

This is the canonical momentum conjugate to the phase θ. Thus the clock evolves and carries a genuine momentum.

The cylinder generator $D = \partial_t$ acts trivially on the stationary metric and rotates the scalar pair, as already distinguished from R and K in (3):

$$\mathcal{L}_D g = 0, \qquad \mathcal{L}_D T = \omega R T. \tag{21}$$

The action-derived covariant presymplectic form then gives the exact identity

$$\Omega_{\text{total}}(\delta, \mathcal{L}_D) = \omega\, \delta Q_R. \tag{22}$$

The pure-Weyl cross term vanishes because $\mathcal{L}_D g = 0$; the matter term is the variation of the Noether charge (20). This establishes `P09-C3`, but by itself it does not decide whether D is gauge. The decision depends on the allowed tangent space.

3.1 The fixed-coupling slice and the lapse normalization

To define that tangent space without gauge-fixing away its constraint, retain the homogeneous lapse and use the common Weyl scale to set $a(t) = 1$ only after variation:

$$ds^2 = -N(t)^2 dt^2 + \sigma_1^2 + \sigma_2^2 + c(t)^2\sigma_3^2, \qquad T = \rho(t)(\cos\theta(t), \sin\theta(t)). \tag{23}$$

With $V_0 = 16\pi^2$, the reduced action is

$$\frac{S_{\text{red}}}{V_0} = \int dt\, Nc \left[\frac{\alpha_B}{8}C^2 + \frac{\dot\rho^2 + \rho^2\dot\theta^2}{2N^2} - \frac{R\rho^2}{12} - \frac{\lambda\rho^4}{4}\right]. \tag{24}$$

For auditability, the curvature scalars before the static specialization are

$$R = -\frac{c^3N^3 - 4cN^3 + 4\dot c\dot N - 4N\ddot c}{2cN^3}, \tag{25}$$

$$C^2 = \frac{4A_-A_+}{3c^2N^6}, \qquad A_\pm = c^3N^3 - cN^3 + \dot c\dot N - N\ddot c \pm 3c\dot cN^2. \tag{26}$$

Define the normalized lapse equation

$$E_N := \frac{1}{V_0}\frac{\delta S_{\text{red}}}{\delta N}. \tag{27}$$

Equations (24)–(27) fix the normalization in the linear identity below.

The open background family is a family across theories, not a modulus of one fixed theory. Eliminating the background variables gives

$$F(q; \alpha_B\lambda) = q^2 - 5q + 1 + 6\alpha_B\lambda(1-4q)^2 = 0, \tag{28}$$

and $\partial_q F$ after substituting the branch value of $\alpha_B\lambda$ is $3(6q-1)/(4q-1)$, which never vanishes on (12). Schematically,

$$\left\{\bar\Phi(q), \alpha_B, \lambda(q)\right\}_{q\in I} \longrightarrow \{(\alpha_B, \lambda)\}, \qquad T_{\bar\Phi}\mathcal{S}_{\alpha_B,\lambda} = \ker D\mathcal{E}_{\bar\Phi} \cap \ker d\alpha_B \cap \ker d\lambda. \tag{29}$$

Thus $\delta N, \delta c, \delta\rho, \delta\theta$ are varied in deriving the equations, but an allowed tangent has $\delta\alpha_B = \delta\lambda = 0$ and must satisfy every linearized equation, including $\delta E_N = 0$.

Proposition 3.1 (Fixed-coupling charge theorem; `P09-C4`)**.** *On the smooth fixed-α_B, fixed-λ linearized solution space about the positive Berger clock background,*

$$\delta Q_R = 0. \tag{30}$$

Hence (4) *holds and D is presymplectically null on this linearized phase space.*

Proof. Direct variation of (24), followed by restriction to the background branch, gives

$$\delta E_N = -\frac{\alpha_B q^{3/2}}{2}\frac{\delta Q_R}{Q_R}. \tag{31}$$

Here

$$\frac{\delta Q_R}{Q_R} = \frac{\delta c}{\sqrt{q}} + 2\frac{\delta\rho}{\rho} + \frac{\delta\dot\theta}{\omega} - \delta N \tag{32}$$

at $a = N = 1$. The coefficient in (31) is nonzero throughout (12), so every homogeneous allowed tangent has $\delta Q_R = 0$. At the rational fixture the same row is

$$\delta E_N = \frac{27\sqrt{10}}{320}\delta N - \frac{9}{16}\delta c - \frac{9\sqrt{10}}{80}\delta\dot\theta - \frac{27\sqrt{10}}{160}\delta\rho = -\frac{27\sqrt{10}}{320}\frac{\delta Q_R}{Q_R}. \tag{33}$$

By Lemma 3.2, the compact spatial isometry group preserves the background and the linearized equations, while δQ_R is an invariant linear functional. If a smooth inhomogeneous solution tangent had nonzero δQ_R, averaging it over the compact group would produce a homogeneous solution tangent with the same nonzero value, contradicting (31). Thus (30) holds on the complete declared smooth tangent space. Equation (22) gives the result. □

Lemma 3.2 (Compact averaging)**.** *Let $G = SU(2)_L \times U(1)_R$ be the compact spatial isometry group of the Berger background and let $\mathrm{Av}_G\,\delta\Phi = \int_G g\cdot\delta\Phi\,d\mu_G(g)$. Then Av_G is a continuous map in the smooth Fréchet topology, commutes with the fixed-coupling linearized equations, and obeys*

$$\delta Q_R(\mathrm{Av}_G\,\delta\Phi) = \delta Q_R(\delta\Phi). \tag{34}$$

Proof. Naturality of the equations and G-invariance of $\bar\Phi$ give $D\mathcal{E}_{\bar\Phi}(g\cdot\delta\Phi) = g\cdot D\mathcal{E}_{\bar\Phi}(\delta\Phi)$; Haar integration therefore preserves the kernel and all linearized constraints. The internal charge is the integral over the closed spatial slice of a G-invariant scalar density, so its differential is a G-invariant continuous linear functional. Haar normalization then gives the displayed equality. Residual-isometry zero modes are retained by the average; none is removed by a spectral projector. □

The compact averaging step is the matter-clock analogue of the familiar linearization-stability constraints on compact Cauchy surfaces [9, 10]; here the exact lapse identity (31) supplies the decisive coefficient.

Remark 3.3 (What is fixed). The interval (12) parametrizes theories through the product $\alpha_B\lambda$; it is not a modulus inside one fixed theory. At fixed couplings the stationary branch does not furnish a freely variable q or Q_R. There is therefore no contradiction between $Q_R > 0$ and $\delta Q_R = 0$ on allowed tangents.

4 Gauge explanation and relational evolution

The phase θ changes along D, but a gauge transformation of the total system need not leave every partial observable fixed. Relational quantities compare another field to the clock. If X is a scalar quantity with $DX \neq 0$, then locally one may evaluate X on the slice $\theta = \theta_0$ or form combinations invariant under the simultaneous shift of X and θ. The background momentum (20) makes this clock slice nondegenerate, while Proposition 3.1 says that the infinitesimal cylinder motion is presymplectically null in the compact fixed-coupling tangent space.

This paper proves that θ is a regular, backreacting clock candidate and that D is null in the specified linearized and truncated homological senses. It does *not* construct a complete observable in the sense of Rovelli–Dittrich [4, 5]: no global gauge-invariant map $X \mapsto X|_{\theta=\theta_0}$, uniqueness domain, or complete-observable Poisson algebra is asserted. The periodicity in Remark 2.2 is one reason to keep that distinction explicit. Relational Maxwell redshift fixtures in the repository are follow-up applications and are not dependencies of Theorem 1.1.

This result is sector dependent. A boundary Hamiltonian, nonzero flux, or a phase space allowing $\delta Q_R \neq 0$ would change the classification of D. Nothing here promotes the compact Berger verdict to asymptotically flat, de Sitter, anti-de Sitter, or bounded regions.

5 The complete causal BV complex

Let $(\mathcal{C}_{54}, q_1)$ denote the gauge-fixed classical BV complex at the rational point (17). Its row degrees have multiplicities

$$5 \text{ in degree } -1, \qquad 22 \text{ in degree } 0, \qquad 22 \text{ in degree } 1, \qquad 5 \text{ in degree } 2. \tag{35}$$

Table 1 summarizes the component implementation. Here τ is the clock-adapted time ghost, σ the Weyl row, and (R, Θ) the dressed amplitude and phase rows.

Degree	Bundle rows	Minimal	Nonminimal
-1	spatial diffeomorphism ghosts c_i, temporal ghost τ, Weyl ghost σ	5	0
0	dressed metric $\widehat{h}_{\mu\nu}$, R, Θ; multipliers b_a; antighost duals $\bar{c}_a^*$	12	10
1	$\widehat{h}^*_{\mu\nu}, R^*, \Theta^*$; multiplier duals b_a^*; antighosts $\bar{c}_a$	12	10
2	ghost/identity antifields c_i^*, τ^*, σ^*	5	0
	Total	34	20

Table 1: The complete 54-component gauge-fixed presentation. The five nonminimal labels a comprise three spatial-diffeomorphism, one temporal and one Weyl row.

The invariant retained complex has four bundle rows

$$\Gamma(T^*S^3)[-1] \longrightarrow \Gamma(S^2T^*M) \longrightarrow \Gamma(S^2T^*M)[1] \longrightarrow \Gamma(T^*S^3)[2], \tag{36}$$

of component ranks $3, 10, 10, 3$. Its unary blocks are the spatial gauge generator, the retained action Hessian and the negative formal-adjoint gauge identity. The remaining eight minimal clock/temporal/Weyl rows and twenty nonminimal rows form the 28-component contractible extension. Thus the number 54 is an all-row completeness audit, whereas the statement $\mathcal{C}_{54} \simeq \mathcal{C}_{26}$ is independent of the chosen component ordering.

A gauge-fermion canonical shear preserves the frozen odd Darboux pairing and gives a support-local cyclic contraction

$$\pi_{\rm cl}\iota_{\rm cl} = \mathrm{id}, \qquad \mathrm{id} - \iota_{\rm cl}\pi_{\rm cl} = q_1 S_{\rm cl} + S_{\rm cl} q_1 \tag{37}$$

onto the retained complex (36). This is claim `P09-C5`.

For compactly supported sources, the retained complex has advanced and retarded degree-(-1) chain contractions

$$\Lambda_{26,\pm} : \Gamma_c(\mathcal{C}_{26}) \longrightarrow \Gamma_{J^\pm}(\mathcal{C}_{26})[-1]. \tag{38}$$

Lifting them through (37) gives

$$\Lambda_{54,\pm} = S_{\rm cl} + \iota_{\rm cl}\Lambda_{26,\pm}\pi_{\rm cl}, \tag{39}$$

and therefore

$$q_1\Lambda_{54,\pm} + \Lambda_{54,\pm}q_1 = \mathrm{id}_{54}, \qquad \mathrm{supp}(\Lambda_{54,\pm}f) \subseteq J^\pm(\mathrm{supp}\, f). \tag{40}$$

Indeed,

$$\begin{aligned}[q_1, \Lambda_{54,\pm}] &= [q_1, S_{\mathrm{cl}}] + \iota_{\mathrm{cl}}[q_{1,26}, \Lambda_{26,\pm}]\pi_{\mathrm{cl}} \\ &= (1 - \iota_{\mathrm{cl}}\pi_{\mathrm{cl}}) + \iota_{\mathrm{cl}}\pi_{\mathrm{cl}} = 1. \end{aligned} \tag{41}$$

All maps in the retract are differential or pointwise algebraic. No inverse Laplacian, inverse curl, transverse–traceless projector, or mode projector is used. The complementary-degree adjoint relation and D-equivariance also lift. These statements comprise `P09-C6`.

The cyclic higher primitives below instead map compactly supported inputs to sections supported in the two-sided causal hull $J = J^+ \cup J^-$. They are neither the causal difference $\Lambda_+ - \Lambda_-$ nor separate advanced/retarded higher operations.

6 Action-derived interaction tensors

For a fluctuation ϕ about $\bar{\Phi}$, expand the classical BV vector field in the suspended graded-symmetric factorial convention,

$$Q_{\bar{\Phi}}(\phi) = q_1(\phi) + \frac{1}{2!}q_2(\phi, \phi) + \frac{1}{3!}q_3(\phi, \phi, \phi) + O(\phi^4). \tag{42}$$

Thus q_n has n inputs, and q_3 is the fourth action derivative paired with one output. If $\langle -, - \rangle_{\mathrm{BV}}$ denotes the degree-one odd Darboux pairing, the operations are defined invariantly by the polarized identity

$$\langle x_0, q_n(x_1, \ldots, x_n)\rangle_{\mathrm{BV}} = \frac{\partial^{n+1}}{\partial t_0 \cdots \partial t_n} S_{\mathrm{BV}}\left[\bar{\Phi} + \sum_{j=0}^{n} t_j x_j\right]\Bigg|_{t=0}, \tag{43}$$

with the displayed ordering interpreted in the frozen suspended Koszul convention. The local clock dressing and gauge-fermion shear are then applied as BV-canonical transformations to every input, output and dual row.

The proof pipeline is therefore

$$S_{\mathrm{BV}} \xrightarrow{\text{mixed variations}} (q_1, q_2, q_3) \xrightarrow{\text{Cartan sources}} A_D^{(1)}, A_D^{(2)}, A_D^{(3)} \xrightarrow{\text{Green/Hom contraction}} \iota_D^{(1)}, \iota_D^{(2)}, \iota_D^{(3)}. \tag{44}$$

The operations q_2 and q_3 are obtained by differentiating the covariant Weyl–clock master action on arbitrary four-dimensional jets and transporting the result through the same clock and gauge-fermion canonical maps used at arity one. They are not fits to residual modes or a harmonic cutoff.

The arity-two and arity-three parts of $Q^2 = 0$ are

$$[q_1, q_2] = 0, \tag{45}$$

$$[q_1, q_3] + \frac{1}{2}[q_2, q_2] = 0. \tag{46}$$

The common action origin supplies cyclicity with respect to the odd Darboux pairing. The stationary invariant-frame generator $D = e_0$ acts linearly, so its higher Taylor components vanish through this order,

$$L_D^{(2)} = L_D^{(3)} = 0, \tag{47}$$

and exact coefficientwise verification gives

$$[D, q_2] = [D, q_3] = 0. \tag{48}$$

Claims P09-C7 and P09-C8 bind these statements to the portable all-row coefficient exports.

The abstract L_∞ relations follow from the classical master equation; the large PBW calculation verifies that the concrete curved coefficients, canonical transports and signs instantiate those relations without a dropped row. As a cross-producer check, an independently written homogeneous reduced action gives eight fourth derivatives in the $(\widehat{h}_{00}, R)$ sector. They agree exactly with all sixteen ordered zero-jet coefficients in the published $\widehat{h}^*_{00}$ and R^* rows. This probes both the lapse and dressed radial/Weyl directions without importing the q_3 producer. It is deliberately not described as a complete second derivation of the payload.

Arity three is the first test involving both q_3 and the composite $[q_2, q_2]$ channel. It is not the end of the classical expansion: inverse metrics, curvature and canonical field redefinitions generally generate nonzero q_n for $n \geq 4$ even though the bare scalar potential is quartic.

7 Causal Cartan contraction through arity three

Write

$$\iota_D = \iota_D^{(1)} + \iota_D^{(2)} + \iota_D^{(3)} + \cdots \tag{49}$$

for a degree-(-1) Cartan homotopy. At unary order,

$$[q_1, \iota_D^{(1)}] = D. \tag{50}$$

A one-sided raw solution follows from (40); averaging it with its convention-correct cyclic adjoint yields a cyclic unary primitive with support in the two-sided causal hull.

For clarity, the chain contraction is used on the multilinear Hom complex, not as an inverse of a coefficient matrix. If $\delta = [q_1, -]$ and A is a homogeneous multilinear operator, set

$$\mathfrak{H}_\pm(A) := \Lambda_{54,\pm} \circ A. \tag{51}$$

The graded derivation rule and $[q_1, \Lambda_{54,\pm}] = 1$ give

$$\delta\mathfrak{H}_\pm(A) + \mathfrak{H}_\pm\delta(A) = A. \tag{52}$$

This identity includes the q_1 action on every input slot; no input term is discarded. It is the step that turns a q_1-closed Cartan source into a raw causal primitive.

At binary order define

$$A_D^{(2)} = [q_2, \iota_D^{(1)}]. \tag{53}$$

Equations (45) and (48) make $A_D^{(2)}$ q_1-closed. A raw causal primitive is cyclicized with $\mathrm{Cyc}_3 = (1 + \tau + \tau^2)/3$, giving

$$[q_1, \iota_D^{(2)}] = -[q_2, \iota_D^{(1)}]. \tag{54}$$

This is P09-C9.

The ternary source is

$$A_D^{(3)} = [q_3, \iota_D^{(1)}] + [q_2, \iota_D^{(2)}] - L_D^{(3)}. \tag{55}$$

Lemma 7.1 (Closure of the ternary source)**.** *The complete arbitrary-input source* (55) *obeys*

$$[q_1, A_D^{(3)}] = 0. \tag{56}$$

Proof. Let $\delta = [q_1, \cdot]$. Using (46), (50) and (54), the only nonzero formal channels in $\delta A_D^{(3)}$ are

$$-\frac{1}{2}[[q_2, q_2], \iota_D^{(1)}] + [q_2, [q_2, \iota_D^{(1)}]] - [q_3, D]. \tag{57}$$

Graded Jacobi gives

$$[q_2, [q_2, \iota_D^{(1)}]] = \frac{1}{2}[[q_2, q_2], \iota_D^{(1)}], \tag{58}$$

and $[D, q_3] = 0$ annihilates the last term. Thus the normalized coefficients are $-1/2 + 1/2 = 0$ and 0 in the two independent channels. $\square$

Because cyclic coderivations are closed under their graded bracket, $A_D^{(3)}$ is cyclic. The causal chain contraction supplies a raw primitive

$$R^{(3)} = -\mathfrak{H}_+(A_D^{(3)}) = -\Lambda_{54,+} \circ A_D^{(3)}, \qquad \delta R^{(3)} = -A_D^{(3)}. \tag{59}$$

Pair the output with the odd Darboux form before cyclicizing. The resulting four-linear tensor carries the standard Koszul C_4 action and the exact Reynolds projector

$$\mathrm{Cyc}_4 = \frac{1}{4}(1 + \tau + \tau^2 + \tau^3). \tag{60}$$

Since $\tau^4 = 1$, $\mathrm{Cyc}_4^2 = \mathrm{Cyc}_4$. Cyclicity of q_1 makes Cyc_4 commute with δ, while cyclicity of the source gives $\mathrm{Cyc}_4 A_D^{(3)} = A_D^{(3)}$. Therefore

$$\iota_{D,\mathrm{cyc}}^{(3)} = \mathrm{Cyc}_4 R^{(3)}, \qquad [q_1, \iota_{D,\mathrm{cyc}}^{(3)}] = -A_D^{(3)}. \tag{61}$$

This proves the ternary part of Theorem 1.1 and `P09-C10`.

The tensorized C_4 convention matters. The actual 54-row pairing has 27 negatively oriented dual slots. Its duality involution, degree-one pairing and odd skew orientation are checked on every row. The C_4 group law is exhausted over all actual degree classes, covering 978,736 admissible degree-zero row quartets with no defect.

7.1 Support statement

The Taylor operations and pairing are support-local. Green contractions and their cyclic adjoints are causal. Consequently

$$\mathrm{supp}\big(\iota_{D,\mathrm{cyc}}^{(3)}(f, g, h)\big) \subseteq J\big(\mathrm{supp}\, f \cup \mathrm{supp}\, g \cup \mathrm{supp}\, h\big). \tag{62}$$

The cyclic primitive is not asserted to be separately retarded or advanced. The one-sided chain contractions (40) retain that stronger property; cyclic averaging generally does not.

8 Interpretation, failures and open directions

The result resolves one narrow version of the clock question. A healthy matter clock can have $Q_R > 0$ while the cylinder generator remains presymplectically null, because the constraint fixes Q_R within one theory and annihilates its pullback differential. The all-row BV calculation shows that this is not only a homogeneous minisuperspace coincidence: the linearized covariant phase space, causal complex and first interaction recurrences give compatible answers.

The following conclusions are not drawn.

1. No all-orders nonlinear Cartan contraction is claimed. Arity four and higher remain separate gates if the complete master action requires them.
2. No constant-rank nonlinear null foliation, integrated gauge quotient, global deparametrization or complete relational observable is constructed.
3. No Hadamard two-point function, microlocal spectrum condition, renormalized composite, anomaly cancellation or quantum master equation is constructed.
4. No positive graviton Hilbert space or interacting probability theorem follows from the positive clock matter.
5. No noncompact or bounded phase space is covered. There D may acquire a boundary charge or flux and cease to be presymplectically null.
6. Relational-redshift fixtures are applications, not part of the main theorem unless their independent propagation and backreaction gates close.

The correct continuation is therefore not to advertise an unrestricted solution of the problem of time. It is to test whether the same Cartan structure survives higher Taylor order, whether the S^1 clock supports well-defined complete observables on controlled domains, and whether the conclusion persists under renormalized Ward identities and physical boundaries.

9 Reproducibility and claim map

The companion file `09-relational-clocks-berger-d-cartan-computational-supplement.tex` contains the focused claim ledger, exact certificate paths, coefficient inventory, replay commands and freeze checklist. Its machine-readable source is `../d_quotient_classical/certificates/PAPER_09_BERGER_CLAIM_TABLE.json`. The verifier checks strict Draft 2020–12 schema validity, certificate hashes, required true and false flags, and occurrence of every claim identifier in the manuscript sources.

The expensive q_3 action expansion is not silently rerun by the paper build. Its content-addressed Tier-2 receipt records the derivation, canonical transport, row-bounded identity replay and independent portable consumer. Fast paper checks import that frozen artifact by hash. A paper release may be marked `THEOREM_FROZEN` only after clean-tree replay and nonlinear-team sign-off; this working draft is not yet so marked.

References

[1] R. Bach, *Zur Weylschen Relativitätstheorie und der Weylschen Erweiterung des Krümmungstensorbegriffs*, Math. Z. **9** (1921), 110–135.

[2] I. A. Batalin and G. A. Vilkovisky, *Gauge algebra and quantization*, Phys. Lett. B **102** (1981), 27–31.

[3] M. Henneaux and C. Teitelboim, *Quantization of Gauge Systems*, Princeton University Press, 1992.

[4] C. Rovelli, *Partial observables*, Phys. Rev. D **65** (2002), 124013.

[5] B. Dittrich, *Partial and complete observables for Hamiltonian constrained systems*, Gen. Relativ. Gravit. **39** (2007), 1891–1927.

[6] C. Goeller, P. A. Höhn, and J. Kirklin, *Diffeomorphism-invariant observables and dynamical frames in gravity: reconciling bulk locality with general covariance*, arXiv:2206.01193.

[7] C. Crnkowic and E. Witten, *Covariant description of canonical formalism in geometrical theories*, in *Three Hundred Years of Gravitation*, Cambridge University Press, 1987.

[8] I. Khavkine, *Covariant phase space, constraints, gauge and the Peierls formula*, Int. J. Mod. Phys. A **29** (2014), 1430009.

[9] A. E. Fischer and J. E. Marsden, *Linearization stability of the Einstein equations*, Bull. Amer. Math. Soc. **79** (1973), 997–1003.

[10] J. M. Arms, J. E. Marsden, and V. Moncrief, *The structure of the space of solutions of Einstein's equations II: several Killing fields and the Einstein–Yang–Mills equations*, Ann. Phys. **144** (1982), 81–106.

[11] A. S. Cattaneo, P. Mnev, and N. Reshetikhin, *Classical and quantum Lagrangian field theories with boundary*, PoS CORFU2011 (2012), 044.

[12] H. Kajiura, *Noncommutative homotopy algebras associated with open strings*, Rev. Math. Phys. **19** (2007), 1–99.

[13] A. S. Arvanitakis, O. Hohm, C. Hull, and V. Lekeu, *Homotopy transfer and effective field theory I: tree level*, arXiv:2007.07942.

[14] D. Fiorenza and M. Manetti, *L_∞ algebras, Cartan homotopies and period maps*, arXiv:math/0605297.

[15] C. G. Callan, S. Coleman, and R. Jackiw, *A new improved energy–momentum tensor*, Ann. Phys. **59** (1970), 42–73.

[16] M. Benini, G. Musante, and A. Schenkel, *Green hyperbolic complexes on Lorentzian manifolds*, Commun. Math. Phys. (2023), doi:10.1007/s00220-023-04807-5.